1. Let $A=\{1,2,3,4,\ldots\ldots\ldots.10\}$ and $B=\{0,1,2,3,4\}$. The number of elements in the relation $R=\left\{\left(a,b\right)\in A\times A: 2(a-b)^2+3(a-b)\in B\right\}$ is __________.

[2023 (06 Apr Shift 1)]

2. In a group of 100 persons 75 speak English and 40 speak Hindi. Each person speaks at least one of the two languages. If the number of persons who speak only English is α and the number of persons who speaks only Hindi is β, then the eccentricity of the ellipse $25\left(\beta^2x^2+\alpha^2y^2\right)=\alpha^2\beta^2$ is

[2023 (06 Apr Shift 2)]

(1) $\frac{\sqrt{119}}{12}$

(2) $\frac{\sqrt{117}}{12}$

(3) $\frac{3\sqrt{15}}{12}$

(4) $\frac{\sqrt{129}}{12}$

3. Let the number of elements in sets A and B be five and two respectively. Then the number of subsets of $A\times B$ each having at least 3 and at most 6 elements is

[2023 (08 Apr Shift 1)]

(1) 752

(2) 782

(3) 792

(4) 772

4. Let $A=\{0,3,4,6,7,8,9,10\}$ and R be the relation defined on A such that $R\{(x,y)\in A\times A: x-y$ is odd positive integer or $x-y=2\}$. The minimum number of elements that must be added to the relation R, so that it is a symmetric relation, is equal to _________

[2023 (08 Apr Shift 1)]

5. Let $A=\{1,2,3,4,5,6,7\}$. Then the relation $R=\{(x,y)\in A\times A: x+y=7\}$ is

[2023 (08 Apr Shift 2)]

(1) an equivalence relation

(2) symmetric but neither reflexive nor transitive

(3) transitive but neither symmetric nor reflexive

(4) reflexive but neither symmetric nor transitive

6. The number of elements in the set $\left\{n\in\mathbb{Z}: \left|n^2-10n+19\right|<6\right\}$ is _______ .

[2023 (10 Apr Shift 1)]

7. Let $A=\{2,3,4\}$ and $B=\{8,9,12\}$. Then the number of elements in the relation $R=\{((a_1,b_1),(a_2,b_2))\in(A\times B,A\times B)$: a_1 divides b_2 and a_2 divides $b_1\}$ is

[2023 (10 Apr Shift 2)]

(1) 36

(2) 24

(3) 18

(4) 12

8. The number of elements in the set $S=\left\{\theta\in\left[0,2\pi\right]: 3\cos^4\theta-5\cos^2\theta-2\sin^6\theta+2=0\right\}$ is

[2023 (11 Apr Shift 1)]

(1) 10

(2) 8

(3) 12

(4) 9

9. An organization awarded 48 medals in event $'A'$, 25 in event $'B'$ and 18 in event $'C'$. If these medals went to total 60 men and only five men got medals in all the three events, then, how many received medals in exactly two of three events?

[2023 (11 Apr Shift 1)]

(1) 15

(2) 21

(3) 10

(4) 9

10. Let $A=\{1,3,4,6,9\}$ and $B=\{2,4,5,8,10\}$. Let R be a relation defined on $A\times B$ such that $R=\{(a_1,b_1),(a_2,b_2)$: $a_1\leq b_2$ and $b_1\leq a_2\}$. Then the number of elements in the set R is

[2023 (11 Apr Shift 2)]

(1) 160

(2) 52

(3) 26

(4) 180

11. The number of relations, on the set $\{1,\ 2,\ 3\}$ containing $(1,\ 2)$ and $(2,\ 3)$ which are reflexive and transitive but not symmetric, is ________.
[2023 (12 Apr Shift 1)]

12. Let $A=\{-4,-3,-2,0,1,3,4\}$ and $R=\{(a,b)\in A\times A : b=|a| \text{ or } b^2=a+1\}$ be a relation on A. Then the minimum number of elements, that must be added to the relation R so that it becomes reflexive and symmetric, is
[2023 (13 Apr Shift 2)]

13. The number of elements in the set $\{n\in\mathbb{N} : 10\le n\le 100 \text{ and } 3^n-3 \text{ is a multiple of } 7\}$ is ________.
[2023 (15 Apr Shift 1)]

14. Let $A=\{1,\ 2,\ 3,\ 4\}$ and R be a relation on the set $A\times A$ defined by $R=\{((a,\ b),\ (c,\ d)) : 2a+3b=4c+5d\}$. Then the number of elements in R is ________.
[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (18)	**2.** (1)	**3.** (3)	**4.** (19)	**5.** (2)	**6.** (6)	**7.** (1)	**8.** (4)
9. (2)	**10.** (1)	**11.** (4)	**12.** (7)	**13.** (15)	**14.** (6)		

1. (18)

Given,

$A=\{1,2,3,\ldots,10\}$ $B=\{0,1,2,4\}$

$(a,b)\in A\times A$ such that

$2(a-b)^2+3(a-b)-k=0$

where $k\in\{0,1,2,3,4\}$

Now finding discriminant $D=9-4\times 2(-k)$

And $9-4\times 2(-k)$ a perfect square for any possible (a,b) as $(a-b)$ will be a integer,

So, $9+8k$ is a perfect square

$\Rightarrow k=0$ or $k=2$

Now for $k=0$,

$2(a-b)^2+3(a-b)=0$

$\Rightarrow(a-b)\left[2(a-b)^2+3\right]=0$

$\Rightarrow a-b=0\Rightarrow(a,b)\in\{(1,1),(2,2)\ldots.(10,10)\}$

$\Rightarrow$Total 10 elements belonging to R.

And, $a-b=\frac{3}{2}$ is not possible

Now for $k=2$,

$2(a-b)^2+3(a-b)-2=0$

$\Rightarrow a-b=-2$ or $a-b=\frac{1}{2}$ (not possible)

Now for $a-b=-2$ possible pair will be,

$\Rightarrow(a,b)\in\{(1,3),(2,4),\ldots.(8,10)\}$

$\Rightarrow$ 8 elements belonging to R

Total number of elements will be$=18$

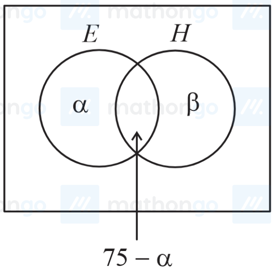
2. (1)

Given,

In a group of 100 persons 75 speak English and 40 speak Hindi,

And each person speaks at least one of the two languages,

And the number of persons who speak only English is α and the number of persons who speaks only Hindi is β,



Now from above diagram we can see that,

$\beta = 100 - 75 = 25$

So, $\alpha = 75 - (40 - 25) = 60$

Hence, number of people who speak English only is $\alpha = 60$ and Hindi is $\beta = 25$,

Now putting the value in given equation of ellipse we get,

$25\left[\frac{x^2}{60^2} + \frac{y^2}{25^2}\right] = 1$

$\Rightarrow \frac{x^2}{36 \times 4} + \frac{y^2}{25} = 1$

Now comparing with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ we get, $a = 12$ & $b = 5$

Now we know that,

Eccentricity of ellipse is given by $e = \sqrt{1 - \frac{b^2}{a^2}}$

$\Rightarrow e = \sqrt{1 - \frac{25}{36 \times 4}} = \frac{\sqrt{119}}{12}$

3. (3)

Given,

The number of elements in sets A and B be five and two respectively,

So, $n(A) = 5$ and $n(B) = 2$

Now Cartesian product will be, $n(A \times B) = 10$

Number of subsets having three elements $= {}^{10}C_3$

Number of subsets having four elements $= {}^{10}C_4$

Number of subsets having five elements $= {}^{10}C_5$

Number of subsets having six elements $= {}^{10}C_6$

So, number of subsets having atleast 3 and at most 6 elements will be $= {}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 + {}^{10}C_6$

$= 120 + 210 + 252 + 210$

$= 792$

4. (19)

Given,

Set $A = \{10, 9, 8, 7, 6, 4, 3, 0\}$

Now relation $x - y$ is odd or $x - y = 2$ can be given by,

$\{R = (10,9), (10,8), (10,7), (10,3), (9,8), (9,7), (9,6), (9,4), (9,0), (8,7), (8,6), (8,3), (7,6), (7,4), (7,0), (6,4), (6,3), (4,3), (3,0)\}$

So, total there are 19 elements and all the elements of R, (a, b) are of type $a > b$.

Hence, we need to add total of 19 more elements to R to make in symmetric

5. (2)

We have, $A = \{1, 2, 3, 4, 5, 6, 7\}$

Reflexive: A relation R on a set A is said to be reflexive if every element of A is related to itself.

Thus, R is reflexive $\Leftrightarrow (a, a) \in R$ for all $a \in A$

$\because (1, 1), (2, 2), (3, 3), \ldots\ldots (7, 7)$ does not satisfy $x + y = 7$

Hence R is not reflexive.

Symmetric: A relation R is symmetric on a set A iff

$(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

$\Rightarrow x + y = 7$

Now on interchanging y and x we get, $y + x = 7$ which is always true for given set,

Hence R is symmetric.

Transitive: A relation R on A is said to be transitive relation iff

$(a, b) \in R$ and $(b, c) \in R$

$\Rightarrow (a, c) \in R$ for all $a, b, c \in A$

Now taking $(a, b) \equiv (3, 4)$ and $(b, c) \equiv (4, 3)$ so $(a, c) \equiv (3, 3)$ does not satisfy $x + y = 7$,

Hence, R is not transitive and not equivalence.

Therefore, R is only Symmetric.

6. (6)

Given,

$|n^2 - 10n + 19| < 6$

Now we know that,

$|x| < a \Rightarrow -a < x < a$

Now using above formula we get,

$\Rightarrow -6 < n^2 - 10n + 19 < 6$

$\Rightarrow n^2 - 10n + 25 > 0$ and $n^2 - 10n + 13 < 0$

$\Rightarrow (n - 5)^2 > 0$ and $\left(n - \left(5 - 2\sqrt{3}\right)\right)\left(n - \left(5 + 2\sqrt{3}\right)\right) < 0$

$\Rightarrow n \in \mathbb{Z} - \{5\}$ and $n \in \{2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8\}$

Hence, $n = 2,\ 3,\ 4,\ 6,\ 7,\ 8$

7. (1)

Given,

$A = \{2, 3, 4\},\ B = \{8, 9, 12\}$

And $a_1 \in A,\ b_2 \in B$

Now finding elements where a_1 divides b_2 we get,

$(a_1, b_2) \in \{(2, 8), (2, 12), (3, 9), (3, 12), (4, 8), (4, 12)\}$

Also given, $a_2 \in A, b_1 \in B$

Now finding elements when a_2 divides b_1, we get,

$(a_2, b_1) \in \{(2, 8), (2, 12), (3, 9), (3, 12), (4, 8), (4, 12)\}$

Hence, Number of relations $= 6 \times 6 = 36$

8. (4)

Given,

$3\cos^4\theta - 5\cos^2\theta - 2\sin^6\theta + 2 = 0$

$\Rightarrow \cos^2\theta[3\cos^2\theta - 5] - 2\sin^6\theta + 2 = 0$

$\Rightarrow (1 - \sin^2\theta)(3 - 3\sin^2\theta - 5) - 2\sin^6\theta + 2 = 0$

$\Rightarrow (\sin^2\theta - 1)(3\sin^2\theta + 2) - 2\sin^6\theta + 2 = 0$

Let $\sin^2\theta = t$

$\Rightarrow (t-1)(3t+2) - 2t^3 + 2 = 0$

$\Rightarrow (t-1)(3t+2) - 2(t^3 - 1) = 0$

We know that $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

$\Rightarrow (t-1)(3t+2) - 2(t-1)(t^2 + t + 1) = 0$

$\Rightarrow (t-1)[3t + 2 - 2(t^2 + t + 1)] = 0$

$(t-1)[2t^2 - t] = 0$

$\Rightarrow t = 0, 1, \frac{1}{2}$

For $\sin^2\theta = 0, \theta = 0, \pi, 2\pi$

$\sin^2\theta = 1, \theta = \frac{\pi}{2}, \frac{3\pi}{2}$

$\sin^2\theta = \frac{1}{2}, \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

$\therefore$ Total solution $= 9$

Hence this is the required option.

9. (2)

Let A, B and C denote the set of men who received medals in even A, event B and event C respectively.

Then, $n(A) = 48$, $n(B) = 25$, $n(C) = 18$, $n(A \cup B \cup C) = 60$ and $n(A \cap B \cap C) = 5$,

Now we know that, $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

$\Rightarrow 60 = 48 + 25 + 18 - n(A \cap B) - n(A \cap C) - n(B \cap C) + 5$

$\Rightarrow n(A \cap B) + n(A \cap C) + n(A \cap C) = 48 + 25 + 18 + 5 - 60 = 36$

Therefore, the number of people who received medals in exactly two of the three sports will be $36 - 3(A \cap B \cap C) = 36 - 15 = 21$.

10. (1)

Given,

$A = \{1, 3, 4, 6, 9\}$ & $B = \{2, 4, 5, 8, 10\}$

And relation is given by,

$R = \{(a_1, b_1), (a_2, b_2) : a_1 \leq b_2$ and $b_1 \leq a_2\}$

Now taking cases for $a_1 \leq b_2$ we get,

$a_1 = 1, \ b_2 \in \{2, 4, 5, 8, 10\} \to 5$ cases

$a_1 = 3, \ b_2 \in \{4, 5, 8, 10\} \to 4$ cases

$a_1 = 4, \ b_2 \in \{4, 5, 8, 10\} \to 4$ cases

$a_1 = 6, \ b_2 \in \{8, 10\} \to 2$ cases

$a_1 = 9, \ b_2 \in \{10\} \to 1$ cases

So, total 16 cases will be there

Now finding cases of $b_1 \leq a_2$ we get,

$b_1 = 2, \ a_2 \in \{3, 4, 6, 9\} \to 4$ cases

$b_1 = 4, \ a_2 \in \{4, 6, 9\} \to 3$ cases

$b_1 = 5, \ a_2 \in \{6, 9\} \to 2$ cases

$b_1 = 8, \ a_2 \in \{9\} \to 1$ cases

So, here total 10 cases,

Hence, total elements in relation $= 16 \times 10 = 160$

11. (4)

Given,

Set $A = \{1, 2, 3\}$

Now Cartesian product $A \times A = \{(1,1),(2,1),(1,2),\ldots\ldots\ldots,(3,3)\}$

Now, given the relation is reflexive,

So, $(1,\ 1),\ (2,\ 2),\ (3,\ 3)\ \in\ R$

Also given $(1,\ 2),\ (2,\ 3) \in\ R,\ (1,\ 3)$ must $\in\ R$

Now finding, **Possible cases** :

Case-1: All of $(2,\ 1),\ (3,\ 2),\ (3,\ 1)\ \notin R \to 1$ relation.

Case-2: Only one of $(2,\ 1),\ (3,\ 2),\ (3,\ 1) \in R \to 3$ relations.

For example if relation is $\{(1,\ 1),\ (2,\ 2),\ (3,\ 3),(2,1)\}$ then it is reflexive as well as transitive as $(2,2),(2,1) \to (2,1)$ is present in relation

Note that exactly two of $(2,\ 1),\ (3,\ 2),\ (3,\ 1) \in R$ is not possible because if two of these $\in R$, third must $\in R$ to make relation transitive.

Total number of relations $= 4$

12. (7)

Given,

$A = \{-4, -3, -2, 0, 1, 3, 4\}$ and $R = \left\{(a, b) \in A \times A : b = |a| \text{ or } b^2 = a + 1\right\}$ be a relation on A,

So, the relation is given by,

$R = \{(-4, 4), (-3, 3), (0, 0), (1, 1), (3, 3), (4, 4), (0, 1), (3, -2)\}$

Now, relation to be reflexive $(a, a) \in R\ \forall\ a \in A$

$\Rightarrow (-4, -4), (-3, -3), (-2, -2)$ also should be added in R.

Now relation to be symmetric if $(a, b) \in R$, then $(b, a) \in R\ \forall\ a, b \in A$

$\Rightarrow (4, -4), (3, -3), (1, 0), (-2, 3)$ also should be added in R

Hence, minimum number of elements to be added to

$R = 3 + 4 = 7$

13. (15)

Given that $n \in [10, 100]$ and $n \in N$.

Let us take the values of $n = 1, 2, 3, \ldots$ and check whether $3^n - 3$ is divisible by 7 or not.

$\Rightarrow 3^1 - 3 = 0$ is divisible by 7

$\Rightarrow 3^2 - 3 = 3$, 3 is not divisible by 7

$\Rightarrow 3^6 = 7k + 1$ is not divisible by 7

$\Rightarrow 3^7 = 7\alpha + 3$,

$\Rightarrow 3^7 - 3 = 7\alpha$ is divisible by 7

Similarly $\Rightarrow 3^{13} = (7k + 1)(7\alpha + 3)$

$= 7\beta + 3$

$\Rightarrow 3^{13} - 3 = 7\beta$ is divisible by 7

If we observe the pattern for $n = 1, 7, 13$, $3^n - 3$ is divisible by 7.

That means the series forms an AP with $d = 6$.

Also $n \in [10, 100]$

The required progression is $13, 19, \ldots\ldots a_n$

$\Rightarrow a_n = 13 + (n - 1)6$

But $13 + (n - 1)6 < 100$

$\Rightarrow n < 15.5$

$\Rightarrow n = 15 \in N$

Therefore, the required answer is 15.

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14. (6)

Given,

Set $A = \{1, 2, 3, 4\}$

And relation on $A \times A$ as $2a + 3b = 4c + 5d$

Now, Maximum value of $2a + 3b = 20$ at $(4,\ 4)$

And Minimum value of $4c + 5d = 9$ at $(1,\ 1)$

So, $4c + 5d$ can be equal to $9,\ 13,\ 14,\ 17,\ 18,\ 19$

Now $2a + 3b$ can be 9 if $(a,\ b) = (3,\ 1) \Rightarrow (c,\ d) = (1,\ 1)$

Similarly, $2a + 3b$ can be 13, if $(a,\ b) = (2,\ 3) \Rightarrow (c,\ d) = (2,\ 1)$

And $2a + 3b$ can be 14, if $(a,\ b) = (4,\ 2)$ or $(1,\ 4)$

$\Rightarrow (c,\ d) = (1,\ 2)$

Similarly, $2a + 3b$ can be 17, if $(a,\ b) = (4,\ 3) \Rightarrow (c,\ d) = (3,\ 1)$

And $2a + 3b$ can be 18 if $(a,\ b) = (3,\ 4) \Rightarrow$

$(c,\ d) = (2,\ 2)$

Hence, there are total 6 elements which satisfy the given relation.